

By-Topic Practice Exercises

MCQ + Data-Based Questions (DSE-style)

Work through each topic. Circle ONE best answer for MCQs. Answers and a full marking scheme (with mark allocation and accepted key terms) are at the END of this booklet. All content is within the S.3 Topic 2 syllabus.

Topic 1 — Structure of the Earth

Part A · Multiple Choice

- Which of the following correctly states the physical state of an Earth layer?
 - Lower mantle — molten
 - Outer core — solid
 - Asthenosphere — plastic / molten
 - Inner core — liquid
- Which combination makes up the lithosphere?
 - The crust only
 - The crust + the entire upper mantle
 - The crust + the uppermost SOLID part of the upper mantle
 - The crust + the asthenosphere
- Layer X is liquid and made mainly of iron and nickel. X is the:
 - asthenosphere
 - lower mantle
 - outer core
 - inner core
- Which of the following lists the three main layers of the Earth in the correct order, from the surface towards the centre?
 - Crust → core → mantle
 - Crust → mantle → core
 - Mantle → crust → core
 - Core → mantle → crust
- The lithosphere and the asthenosphere are distinguished mainly by their:
 - chemical composition
 - physical state (solid vs plastic)
 - colour
 - depth below sea level only

Part B · Data-Based Question

Table 1 shows information about four layers of the Earth, labelled P–S (not in order).

Layer	Physical state	Note
P	Solid	Continental (thick) & oceanic (thin) types
Q	Plastic / molten	Plates float and move on it
R	Liquid	Made of iron & nickel
S	Solid	Made of iron & nickel

- Name layers P, Q, R and S. **(4 marks)**
- Which layer allows tectonic plates to move? Explain your answer. **(2 marks)**

(c) State ONE piece of evidence from the table that R is the outer core, not the inner core. **(1 mark)**

Topic 2 — Plate Tectonics & Plate Boundaries

Part A · Multiple Choice

- Plates move apart at a plate boundary acted on by which force?
 - Compressional force
 - Tensional force
 - Shearing force
 - Gravitational force
- The San Andreas Fault is an example of which boundary, between which two plates?
 - Convergent; Pacific & Eurasian
 - Transform; Pacific & North American
 - Divergent; North American & Eurasian
 - Convergent; Pacific & North American
- What drives the movement of tectonic plates over the asthenosphere?
 - The Earth's rotation
 - Mantle convection currents
 - Ocean surface currents
 - The gravitational pull of the Moon
- Which row correctly matches boundary – force – alternative name?
 - Divergent – compressional – constructive
 - Convergent – tensional – destructive
 - Transform – shearing – conservative
 - Divergent – shearing – destructive
- The Mid-Atlantic ridge forms at a ____ boundary, which is mostly found ____.
 - convergent; on land
 - divergent; under the ocean
 - transform; under the ocean
 - divergent; on land
- Which of the following is NOT a tectonic plate?
 - Nazca
 - Cocos
 - Scotia
 - Andes

Part B · Data-Based Question

Figure 2 shows two plates, X and Y, with arrows indicating they are moving TOWARDS each other. A mountain belt lies along their boundary.

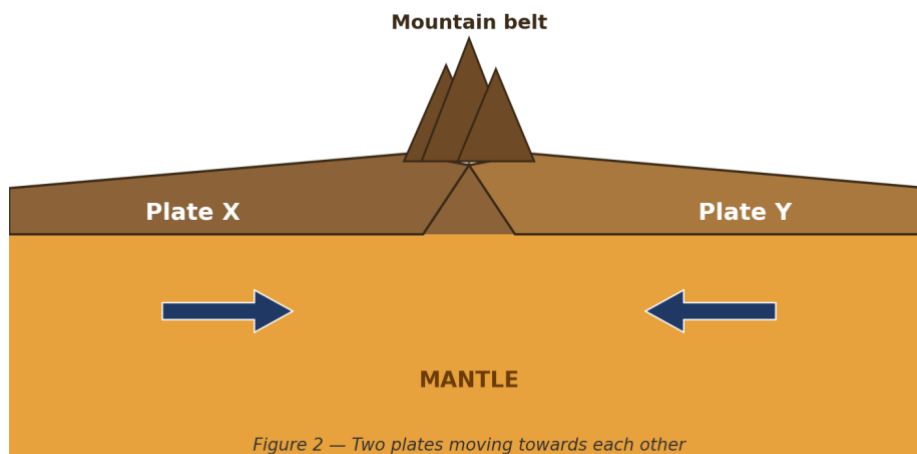


Figure 2 — Two plates moving towards each other

- (a)** Name the type of plate boundary between X and Y. **(1 mark)**
- (b)** State the force acting at this boundary and give ONE alternative name for it. **(2 marks)**
- (c)** With reference to plate movement, explain how this boundary leads to the formation of an earthquake. **(3 marks)**

Topic 3 — Earthquakes: Formation & Description

Part A · Multiple Choice

1. The focus of an earthquake is best described as the:
 - A. point on the surface above where energy is released
 - B. underground point where energy is released
 - C. point of the strongest shaking, always at 0 km depth
 - D. border between two plates
2. A shallow-focus earthquake is usually MORE destructive than a deep-focus one because:
 - A. it always has a higher magnitude
 - B. less energy is lost before the seismic waves reach the surface
 - C. it is measured on the Modified Mercalli scale
 - D. its epicentre is further away
3. Approximately how many times more energy is released by an M8.0 earthquake than an M6.0 earthquake?
 - A. about 2 times
 - B. about 100 times
 - C. about 1 000 times
 - D. about 31.6 times
4. Magnitude differs from intensity because magnitude:
 - A. varies from place to place
 - B. is measured on the Modified Mercalli scale
 - C. is fixed for all places and measures energy released from the focus
 - D. measures the degree of damage at one location
5. Which scale measures the degree of damage at a particular place?
 - A. Richter scale
 - B. Modified Mercalli scale
 - C. Beaufort scale
 - D. Saffir–Simpson scale
6. Earthquakes are LEAST likely to occur:
 - A. at convergent boundaries
 - B. at transform boundaries
 - C. far away from any plate boundary
 - D. along the Pacific Ring of Fire

Part B · Data-Based Question

Table 3 shows two earthquakes, A and B, that struck the same region.

Earthquake	Magnitude (Richter)	Focal depth	Distance of nearest city from epicentre
A	M6.5	15 km	20 km
B	M6.5	250 km	25 km

- (a) Define the term 'magnitude'. (1 mark)
- (b) With reference to the data, suggest which earthquake is likely to cause GREATER destruction. Give TWO reasons. (4 marks)
- (c) Briefly outline how an earthquake forms at a transform plate boundary. (3 marks)

Topic 4 — Impacts of Earthquakes & Measures

Part A · Multiple Choice

- Which of the following is a LONG-term effect of an earthquake?
 - Fire caused by gas leakage
 - Collapse of buildings
 - Bad psychological effect on victims
 - A tsunami
- Japan's 'J-Alert' emergency system is an example of which measure?
 - Hazard-proof structures
 - Monitoring & warning systems
 - Land-use planning
 - Post-hazard financial resources
- Drawing up hazard-zone maps and keeping open areas as shelters belongs to which measure?
 - Education & drills
 - Land-use planning
 - Rescue systems
 - Monitoring & warning
- Which measure most directly reduces the number of deaths AFTER an earthquake strikes?
 - Taking out insurance
 - A well-trained rescue team with life detectors
 - Laws limiting building height
 - Regular public drills
- Which of the following is NOT a short-term effect of an earthquake?
 - Landslides
 - Spread of disease
 - Obstruction of long-term economic development
 - Fire

Part B · Data-Based Question

A magnitude M7.2 earthquake strikes a densely-populated city. Below are some reported effects.

#	Reported effect
1	Roads and bridges damaged, hindering rescue
2	Victims suffer long-term trauma and anxiety
3	Gas pipes break and fires spread
4	Economic development of the region is obstructed for years

- Classify effects 1–4 into short-term effects and long-term effects. (2 marks)
- Suggest TWO measures the government could take BEFORE a hazard to reduce such damage, and explain how each works. (4 marks)
- Name the SIX general types of measure used to cope with earthquakes (the M-B-P-E-R-F list). (3 marks)

Topic 5 — Level of Destruction (incl. MDCs vs LDCs)

Part A · Multiple Choice

- Two earthquakes have equal magnitude. The one causing GREATER loss of life most likely occurred:
 - in a low-density rural area of an MDC at noon
 - in a high-density city of an LDC at midnight
 - in an MDC where regular drills are held
 - offshore, far from any population
- Which is the BEST single reason why LDCs usually suffer more than MDCs from similar earthquakes?
 - Earthquakes always have higher magnitude in LDCs
 - LDCs lack capital and technology for warning and rescue
 - Earthquakes in LDCs always have a shallower focus
 - LDCs are always closer to plate boundaries
- Which factor has a NEGATIVE relationship with destruction (more of it → LESS loss)?
 - Population density
 - Magnitude
 - Preparedness / awareness of people
 - Duration of shaking
- An earthquake that strikes at midnight tends to cause more deaths because:
 - magnitude is higher at night
 - people have lower awareness and are unprepared
 - seismic waves travel faster in the dark
 - rescue teams respond faster at night
- Destruction tends to be GREATEST when the affected place is:
 - far from the epicentre with a deep focus
 - close to the epicentre with a shallow focus
 - far from the epicentre with a shallow focus
 - close to the epicentre with a deep focus

Part B · Data-Based Question

Table 5 shows two countries struck by earthquakes of similar magnitude (about M7.0).

	Country J	Country N
Level of development	MDC	LDC
Typical building type	Earthquake-proof design	Stone, brick & wood
Warning / drills	J-Alert; regular drills	None
Reported deaths	About 200	About 90 000

- Define 'magnitude' and explain why it is described as 'fixed for all places'. (2 marks)
- With reference to the table, explain why Country N suffered far more deaths than Country J. (4 marks)

Topic 6 — Weather, Climate & Tropical Cyclones

Part A · Multiple Choice

- Weather differs from climate in that weather refers to:
 - the average conditions of a place over about 30 years
 - conditions of the lowest atmosphere over a SHORT period
 - temperature only
 - conditions over a long period
- A tropical cyclone (typhoon) is best described as:
 - an intense high-pressure system
 - an intense low-pressure system
 - a long spell of dry, sunny weather
 - a warm front bringing light rain
- The strength of a tropical cyclone is classified mainly according to its:
 - maximum sustained wind speed near the centre
 - air temperature
 - total rainfall
 - distance from land
- Which is the correct SI-style unit for the matching element?
 - Rainfall – hPa
 - Air pressure – mm
 - Wind speed – m/s
 - Relative humidity – °C
- A wind speed of 20 m/s is equal to how many km/h? ($\times 3.6$)
 - 5.6 km/h
 - 36 km/h
 - 72 km/h
 - 20 km/h
- In Hong Kong, tropical cyclones are MOST frequent in:
 - January – February
 - August – September
 - November – December
 - March – April

Part B · Data-Based Question

- (a) Complete the table by writing the correct unit for each weather element. (3 marks)

Weather element	Unit
Rainfall	_____
Air pressure	_____
Temperature	_____

- (b) State the difference between weather and climate. (2 marks)
- (c) In Hong Kong, in which months do tropical cyclones occur, and when are they most frequent? (2 marks)

ANSWERS & MARKING SCHEME

Each accepted point = 1 mark unless stated. Bracketed figures show the marks available. Award marks for any valid alternative wording carrying the same geographical meaning.

Topic 1 — Structure of the Earth

MCQ

Q	Ans	Key reason / accepted term
1	C	Only the asthenosphere (upper mantle) is plastic/molten; lower mantle is solid
2	C	Lithosphere = crust + uppermost SOLID part of upper mantle
3	C	Outer core = liquid, made of iron + nickel
4	B	Order from surface to centre: crust → mantle → core
5	B	Defined by physical state, not chemical composition

DBQ — marking points

Q	Accepted answer	Marks
(a)	P = Crust; Q = Asthenosphere (upper mantle); R = Outer core; S = Inner core	4
(b)	Q / asthenosphere (1) – it is plastic / molten so plates float and move on it (1)	2
(c)	R is LIQUID, whereas the inner core (S) is solid — the outer core is the liquid layer	1

Topic 2 — Plate Tectonics & Boundaries

MCQ

Q	Ans	Key reason / accepted term
1	B	Divergent = tensional force, plates move apart
2	B	San Andreas Fault = transform; Pacific & North American plates
3	B	Mantle convection currents (hot rises, cool sinks)
4	C	Transform – shearing – conservative
5	B	Divergent boundary; mostly under the ocean (e.g. Iceland)
6	D	'Andes' is a mountain range, not a plate

DBQ — marking points

Q	Accepted answer	Marks
(a)	Convergent (destructive) plate boundary	1
(b)	Compressional force (1); also called a destructive boundary (1)	2
(c)	Convection currents converge / drag plates X & Y to collide (1) → stress builds, held by friction (1) → stress exceeds rock strength, rock breaks & releases energy as seismic waves → earthquake (1)	3

Topic 3 — Earthquakes

MCQ

Q	Ans	Key reason / accepted term
1	B	Focus = underground point of energy release (has depth)
2	B	Shallow focus → less energy lost → more destructive

3	C	Each +1 = $\times 31.6$; +2 $\approx 31.6 \times 31.6 \approx 1\ 000$ times
4	C	Magnitude = energy at focus, fixed for all places (Richter)
5	B	Modified Mercalli measures degree of damage (intensity)
6	C	Earthquakes rare far from plate boundaries

DBQ — marking points

Q	Accepted answer	Marks
(a)	Magnitude = the amount of energy released from the focus (Richter scale)	1
(b)	Earthquake A (1). Reasons (any 2 = 2, +1 link to data, max 4 wording): shallower focus 15 km vs 250 km $\rightarrow$ less energy lost; nearer city / shorter horizontal distance 20 km; shallow focus = stronger shaking felt at surface	4
(c)	Two convection currents flow past each other (1) $\rightarrow$ plates slide past horizontally under shearing force (1) $\rightarrow$ stress > strength, rock breaks, energy released as seismic waves (1)	3

Topic 4 — Impacts & Measures

MCQ

Q	Ans	Key reason / accepted term
1	C	Psychological effect = long-term; A/B/D are short-term
2	B	J-Alert = monitoring & warning system
3	B	Hazard-zone maps & open shelters = land-use planning
4	B	Rescue systems (life detectors, trained teams) reduce deaths
5	C	Obstruction of long-term development = long-term effect

DBQ — marking points

Q	Accepted answer	Marks
(a)	Short-term: 1 & 3. Long-term: 2 & 4. (1 mark each pair)	2
(b)	Any 2 measures $\times$ (state + explain = 2 each): hazard-proof structures (resist collapse); land-use planning (avoid risky areas / shelters); monitoring & warning (early evacuation); education & drills (raise awareness)	4
(c)	Monitor $\cdot$ Build (hazard-proof) $\cdot$ Plan (land-use) $\cdot$ Educate $\cdot$ Rescue $\cdot$ Fund (post-hazard finance). Any 6 = 3 marks ($\frac{1}{2}$ each)	3

Topic 5 — Level of Destruction

MCQ

Q	Ans	Key reason / accepted term
1	B	High density + LDC + midnight (low awareness) $\rightarrow$ most deaths
2	B	LDCs lack capital & technology for warning/rescue
3	C	Preparedness $\uparrow \rightarrow$ loss $\downarrow$ (negative relationship)
4	B	Midnight $\rightarrow$ low awareness, people unprepared
5	B	Close epicentre + shallow focus $\rightarrow$ greatest destruction

DBQ — marking points

Q	Accepted answer	Marks
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(a)	Magnitude = energy released from the focus (1); measured on the Richter scale, so it is the same value wherever it is recorded → fixed for all places (1)	2
(b)	Country N is an LDC: weak buildings (stone/brick/wood) collapse easily; no warning system / no drills → low awareness; less capital & technology for rescue; (Country J = MDC with earthquake-proof buildings, J-Alert & drills). Any 4 valid points	4

Topic 6 — Weather, Climate & Cyclones

MCQ

Q	Ans	Key reason / accepted term
1	B	Weather = lowest atmosphere over a SHORT period
2	B	Cyclone = an intense LOW-pressure system
3	A	Cyclone strength classified by maximum sustained wind speed
4	C	Wind speed in m/s (or km/h); rainfall mm, pressure hPa, humidity %
5	C	$20 \times 3.6 = 72$ km/h
6	B	HK cyclones most frequent Aug–Sep

DBQ — marking points

Q	Accepted answer	Marks
(a)	Rainfall = mm; Air pressure = hPa; Temperature = °C (or °F / K)	3
(b)	Weather = conditions of the lowest atmosphere over a SHORT period (1); Climate = average weather of a place over a LONG period, e.g. ~30 years (1)	2
(c)	Occur May–November (1); most frequent in August–September (1)	2